

Teaching Statement

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Mathematics has always been a social, communal human project; I see no real distinction between practicing mathematics and teaching mathematics. I believe that the kinds of communication and reasoning skills taught in mathematics are of broad utility to all university students in their studies and future careers. That diversity of purpose is a great challenge I face as an instructor—I work to teach in a way that emphasizes the universal virtues of rigorous deductive reasoning and precise communication, while adapting to the wide range of goals, backgrounds, and aspirations students can have for their learning. There is no one right way to learn math, and correspondingly, there is no one right way to teach math. I consider the ongoing endeavor for excellence in my teaching to be a significant and rewarding puzzle.

Adaptive, Active, Inquiry-Based Methods

My style of teaching is informed and inspired by non-traditional classroom settings. Some of my earliest mathematical experiences as both a student and instructor have been at summer math programs for motivated high-school students (namely, HCSSiM and MathILy). As an instructor at these programs, I design, plan, and lead creative and open-ended inquiry-based mathematical discussion 1.5–2 hours-a-day, 6 days-a-week, with no assessments or grades of any kind. These experiences are formative for my ideas about pedagogy: every mathematical idea is best motivated, understood, and remembered through a search to discover it.

Moreover, I consider it my goal as an educator to integrate inquiry-based methods and philosophy into the structural necessities of university academics: into courses that I lecture, research projects that I mentor, and discussion sections that I teach. Throughout, my guiding philosophy is that students practice mathematics as *action* rather than as *knowledge*.

In Summer 2023 I taught a course on linear algebra and differential equations. Once each week, I chose to dedicate a part of the lecture time to a mandatory rotation of short student expositions of problems from homework. This had a remarkably empowering effect: after overcoming the mild social pressure to take their turn at the front of the classroom, students were reliably more engaged and more willing to be active participants during my lectures. In the final week, I offered students nominal extra credit if they gave a final presentation on an application of linear algebra to something related to their academic interests. Nearly every

student prepared a presentation, and we saw students engaged in developing their own ideas about how to bring their new skills to bear across interests in science, medicine, music, and more.

In Summer 2025, I mentored an REU program for six undergraduates on a project in geometric representation theory. In order to prepare the students, I designed a curriculum restructuring chunks of the fundamental theory into a form requiring only linear algebra and basic group theory. The curriculum was centered on a sequence of daily open-ended problems¹ which led the students through the process of discovering, on their own terms and in their own language, fundamental facts about matrix groups that are usually understood in terms of sophisticated geometry. We discussed their progress on the problems in class each morning, and I wrote a new problem set at midday for them to work on in the afternoon. Problems usually spun off from questions arising in class. “Can you prove the conjecture that Alice made? Does Bob’s algorithm generalize to this setting?” Crucially, students were tasked, even when learning “standard material,” with finding their own framework for organizing the patterns they observed in examples. At some point, the problems became larger and more research-oriented, but the structure of the work remained the same. By the end of the summer, the students had developed their own framework for understanding topics well beyond an undergraduate level.

In my discussion sections, to complement lectures, I consistently center student presentation and sometimes longer-form projects over instructor explanations. I often ask students to attend prepared to present on some part of classroom material and structure my sections around work in small groups. Where lectures provide clean conceptual material, I structure discussion sections to offer students space and time to productively get their hands dirty playing with mathematical concepts. I consider my consistently high student evaluations evidence that my students feel empowered rather than distressed by my demands for their active participation.

Approachability, Mentorship, and Support

As an instructor and an academic, I work to cultivate an environment in which students feel encouraged to reach out, come to office hours, and participate in mathematics as conversation. I strive to recognize and validate the spectrum of goals and expectations that students bring to my courses; I see this as a matter of foundational respect. I have found

¹Available at <https://math.berkeley.edu/chd/REU/index.html>.

that, sometimes, having a candid non-mathematical conversation about what a student is hoping to achieve in my course is the most impactful thing I can do to support them.

Cultivating an accessible environment requires active humility, breaking down barriers of academic seniority and inscrutability. It requires the far-from-automatic skill of speaking mathematics not only the way I understand it, but from a range of angles and points of view. It requires knowing when my contributions are actually helpful, and when to step back rather than say more. This kind of flexibility of perspective has been especially important in my work this semester teaching for a partnership with the National Educational Equity Lab. The program is offering UC Berkeley's precalculus course remotely to high school students from historically underserved communities, and tests my ability to instruct in a way accessible to those not already embedded in a university environment. In bridging this gap, I am finding that the most crucial work is diagnostic: I have begun to ask my students to prepare 2-3 written questions on course material in advance each week. The writing process gives students time to reflect and build a sense of ownership over the course material, but I also use the questions as springboards for a discussion that starts on the students' terms. The way in which a student phrases a question often reveals as much as the intended question about areas of confusion. Correspondingly, my lesson plans are loose and reactive, with much time spent challenging both the question-asker and other students to walk through examples and problems I generate in response to their apparent misunderstandings. I believe this reactivity makes my classes more inclusive and accessible, and I have noticed consistently broader student engagement as students fall into discussion with each other and actively improve each other's understanding.

Within the university system, I have consistently gone beyond standard instruction to support and mentor undergraduates. In addition to the REU mentorship discussed above, I have mentored directed reading programs in 9 semesters, including one which extended into a for-credit course on algebraic geometry for several students. I am privileged to think that my outreach efforts are valuable to undergraduates: I am one of few graduate students invited by the undergraduate student organization to give seminar talks alongside senior faculty, and several students I have mentored are now PhD students in mathematics.